

MARIOS GLYTSOS

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FIELDS OF RESEARCH INTEREST

Digital Signal Processing, Audio/Audio-Visual Machine Learning, Multimodal Machine Perception, Representation Learning, Music Information Retrieval

PUBLICATIONS

Glytsos, Filntisis, George Retsinas, and Petros Maragos, “Category-Level 6D Object Pose Estimation in Agricultural Settings Using a Lattice-Deformation Framework and Diffusion-Augmented Synthetic Data,” in Proc. of the 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2025.

Garoufis, **Glytsos**, Chourdaki, Filntisis Panagiotis and Maragos Petros. “Power in Unity: Combining in-Domain and out-of-Domain Pre-Training Strategies for EEG-Based Person Identification”, In Proc. IEEE International Conference on Acoustics, Speech, and Signal Processing 2025

Glytsos, Garoufis, Zlatintsi Nancy and Maragos Petros. “Classical Guitar Duet Separation using GuitarDuets – a Dataset of Real and Synthesized Guitar Recordings”, In Proc. *International Society for Music Information Retrieval (ISMIR) 2024*

EDUCATION

09/2017 – 05/2024	National Technical University of Athens – Athens, Greece <i>Integrated MEng in Electrical and Computer Engineering</i> <i>Master Thesis: Music Source Separation on Classical Guitar Duets</i> GPA: 9.25
09/2011 – 07/2021	Music Studies – Athens, Greece <i>Diploma of classical guitar studies,</i> <i>Judges: E. Assimakopoulos, N. Mavroudis, K. Grigoreas</i> Grade: Excellence, A' award
09/2011 – 06/2017	Model High School “Evangeliki School of Smyrna” – Athens, Greece Grade: 19.4/20

WORK EXPERIENCE

09/2024 – Present	Athena Research Center – Athens Project: euROBIN, SynthPose, funded by European Union's Horizon Framework <i>Research Focus: Synthetic data generation for 6D object pose estimation using Generative AI techniques. Development of deep learning models to enhance object detection and pose estimation accuracy, integrating photorealistic synthetic datasets. Contributed to real-time robotics applications, optimizing model performance for deployment on robotic platforms like TIAGo.</i>
02/2024 – 08/2024	B' Radiology Laboratory Medical School of the National and Kapodistrian University of Athens (NKUA) – Athens <i>Research Focus: Development and evaluation of an artificial intelligence algorithm for the estimation of re-computation of treatment plans in image-guided radiotherapy techniques, ANTENA.</i>
07/2022 – 09/2022	DEEPLAB Internship – Athens <i>Research Focus: Visual Common Sense Reasoning</i>

STARTUP EXPERIENCE

05/2022 – Present	Co-Founder of CAIUS: CaiusPaths Project – Metsovo, Greece Project Details: <i>Artificial Intelligence system that exploits satellite data and geospatial information to map forests, identify uncharted hiking trails in nature</i>
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and find solutions for their sustainability.

Role: *Head of Artificial Intelligence, Satellite Data Processing, App Development*

Achievements:

- Forbes 30 under 30, Greek List
- Selected by the European Space Agency Business Incubation Center (ESA BIC) in Greece, receiving a funding of 50,000 euros
- Awarded by the EU Agency for the Space Programme (EUSPA) and Cassini Challenges as one of the top innovative ideas for 2024 leveraging EU space data
- 2nd Place at the Panhellenic Smart Forest Innovation Challenge 2022
- Green Innovation Award in Cross-Fertilisation prizes, part of the CREATHRIV-EU project
- Mentoring teams in competitions: 5th Cassini Hackathon Space for Defense & Security, Greek 3rd Social Hackathon, Apps4Athens Hackathon

ACHIEVEMENTS

2nd & 3rd Place in 2nd Cadenza Challenge, Task 2, 2025 (T040, T019)

3rd Place in Grand Challenge 5 EEG-Music Emotion Recognition of ICASSP 2025 (NTUA-IRAL Team)

1st Place at the European CASSINI HACKATHON 2022 (Mind The Path Team)

1st prize at the Competition of guitar ensemble at 10th Guitar Festival in Palaio Faliro

Distinction in mathematics competition Archimedes 2017

1st prize at the Competition of guitar duets at 7th Guitar Festival in Palaio Faliro

24th prize at Panhellenic physics high school competition

TECHNICAL SKILLS

Programming Languages: *Python (PyTorch, TensorFlow, OpenGL, Pyrender), C, C++, Java, JavaScript, SQL*

Music: *Logic Pro, Reaper, FL Studio, Cubase, Muse Score, iZotope RX8*

Frameworks: *Juce, Django, FastAPI, React, Flutter*

Cloud Computing: *AWS (Sagemaker, EC2 Instances, S3 Buckets, Lambda Functions)*

ORAL PRESENTATIONS

World Trails Conference 2022, ISMR Trails and Technology Presentation

Awarding Event of ESA BIC Greece, Presentation of Caius and its solutions

SOCIETIES AND AFFILIATIONS

Participant as active member at 4th (2015), 5th (2016), 7th (2018), 9th (2020), 10th (2022) Guitar Festival at Palaio Faliro

Member of a Band (Polite Blue)

Evangeliki School of Smyrna Graduates Mathematics Group

Evangeliki School of Smyrna Graduates Drama club

LANGUAGES

English: *Proficient - C2 Level, TOEFL (December 2023)*

Greek: *Native*